Group 2: Data Management Plan

Project Title: NLP Based Smart Mock Interview Assistant: A System to assess Technical and Behavioural interview response.
Team GitHub repository: https://github.com/ekanth3128/SMART-MOCK-INTERVIEW.git

1. Project and Data Overview

The plan explains how the team will gather, document, keep, control and ethically manage data that is used for the Smart Mock Interview Assistant. For technology job-related interviews, questions will be presented, analysis of the typed answers, get score of the answers using NLP models and provide customized feedback.

Research question: Can neural NLP models accurately evaluate technical and behavioural interview responses and provide personalised feedback in a mock interview simulator for technology-related roles?

Area	Plan
Main data	Consistent datasets of the answers to interview questions from Hugging Face and Kaggle, subject to licence checking. .
Supplementary source	Tagged AI Engineer guide may support topic coverage, but will not be copied into a structured dataset unless permission or terms allow it.
Data output	A standardised CSV/JSON question bank with fields id, domain, difficulty, question and reference_answer. The current cleaned output contains 468 records in data/processed.
Testing data	Any user typed responses collected for evaluation will be anonymised, stored separately and deleted when no longer required.

Table 1. Data Collection and Usage Plan

2. Dataset Summary

Dataset / Source	Type	Planned Use	Status
Python Interview Questions 2025	Structured dataset	Main source for Python-focused technical interview questions for the mock interview assistant. It will support technical question coverage for programming-related interview practice.	Selected source used in the raw data folder and processed through the Week 4 cleaning pipeline.
SQL Interview Preparation	GitHub repository / question bank	Source for SQL interview questions, SQL fundamentals and SQL case study materials. It will support database and SQL-related technical interview questions.	Selected source used in the raw data folder and processed through the Week 4 cleaning pipeline.
Data Science, ML, DL and NLP Interview Questions	GitHub repository / question bank	Source for Data Science, Machine Learning, Deep Learning and NLP interview questions and notes. It will support wider technical coverage for data and AI-related roles.	Selected source used in the raw data folder and processed through the Week 4 cleaning pipeline.
Non-technical Interview Questions	GitHub repository / question bank	Source for HR, behavioural, teamwork, leadership, emotional intelligence and diversity-related interview questions. It will support the behavioural interview section of the mock interview assistant.	Planned for later behavioural coverage; not part of the current cleaned technical question bank.

Table 2. Project Dataset Overview

The resultant cleaned dataset is used for the question bank, comparison with reference answers, role/domain filtering and model evaluation. It does not include labelled candidate response scores, so the project will use a controlled rubric or manually reviewed evaluation set to score experiments.

3. Metadata Dictionary

Field	Type	Description / Example
id	String	Unique processed record ID, e.g. 1.
source_name / source_url	String	Original dataset/source and URL for traceability.
domain	String	Question domain, e.g. Python, SQL, ML, DL, NLP, DATA_ANALYSIS or PREDICTIVE_MODELLING.
difficulty	String	Difficulty label, usually Easy, Medium or Hard. One blank difficulty value remains documented in EDA outputs.
Difficulty	String	Easy, Medium, Advanced, or original dataset level.
question	Text	Interview question shown to the user.
reference_answer	Text	Expected answer, marking guide, or key concepts.
answer_type	String	Technical, behavioural, HR, scenario-based.
Keywords	List/Text	Expected concepts, e.g. overfitting, validation, generalisation.
license_status / date_accessed	String/Date	Licence confirmed, pending, restricted, and access date.
processing_notes	Text	Cleaning steps, duplicate handling, standardisation notes.

4. GitHub Document Control and File Naming

GitHub will be used for version control, document storage, weekly updates, code review and dataset documentation. The repository may be private during development if required by the module or if any user-response testing data is stored.

Folder	Purpose
/docs	Planning documents, meeting notes, weekly updates and final reports.
/data/raw	Original downloaded files. These must remain unchanged.
/data/processed	Cleaned and transformed datasets used for modelling.
/data/metadata	Metadata dictionary, source log, licence log and data cleaning notes.
/notebooks	EDA, cleaning, experiments and evaluation notebooks.
/src	Preprocessing code and Week 5 pretrained model code, including src/models/pretrained_sentence_bert.py.

Table 4. Project Repository Organisation

5. README Plan

The title of the project, a brief project statement and research question.

Team roles and responsibility list, Data Lead, GitHub Document Manager.

Report source URL for the dataset, date of source, licence conditions and allowed use of the source.

This document provides instructions for running notebooks, models and the app, as well as the structure of the repository.

Data Management Notes: Raw data is not altered. Processed data is documented in data/processed, and current EDA outputs record one blank difficulty value plus text-quality issues that should be reviewed during later modelling.

Ethics and Privacy: Notes on public datasets, anonymisation and restriction on personal data collection.

6. Security and Storage

The raw data should be stored in /data/raw and not be directly “edited”. Cleaned data is saved to /data/processed and meta data/source logs are saved in /data/metadata. Only authorised team members will be able to access the GitHub repository. The team will not be gathering the direct identifiers including names, emails, student numbers, addresses or phone numbers. User typed data for testing purposes will be anonymised, stored separately, only used for purposes of the project, and then removed after it is no longer needed.

Where applicable, the project will adhere to the principles of the UK GDPR, such as purpose limitation, storage limitation, integrity/confidentiality, data minimisation and accountability.

7. Ethical Statement

Ethical question	Project response
1. Does the data come under GDPR requirements?	Whereas the data collected would fall within GDPR requirements. Data from the public interview question answer should not include any personal data if it is informational in nature consisting of only general questions, answers, subjects and difficulty ratings. If the team type answers from real users during the test, then the team may be affected by GDPR. The project will therefore limit the direct identifiers, anonymise or delete testing responses wherever possible.
2. Does the project conform to UH ethical policies?	If only publicly available and non-personal data sets are used, the project will carry a low risk. If the team is seeking students' answers or those of other participants, the supervisor should have the approval of conducting research on the university or have his answers assessed for low risk status before recruitment or data collection. .
3. Do we have permission to use the data?	Only after checking and logging the licence and terms on Hugging Face and Kaggle, these platforms will be used as primary datasets. If you have permission and/or terms that specifically state that the document can be converted to a structured dataset, the Taggd sifter will be used to augment it.The Taggd sifter will supplement the Taggd guide, unless permission and/or terms explicitly provide otherwise.
4. Are we assured the original data was collected ethically?	The team cannot guarantee this to the fullest extent without documentation from the original source. Review of Dataset cards, Kaggle metadata, terms of use and source notes. Clear, restrictive or inappropriate data sources will not provide datasets to train the model. .

Table 5. Ethical and GDPR Considerations

8. References

Khetan, Y. (2025) *Python Interview Questions 2025*. Kaggle. Available at: https://www.kaggle.com/datasets/yamankhetan/python-interview-questions-2025 (Accessed: 29 May 2026).

Mistry, R. (n.d.) *SQL Interview Preparation*. GitHub. Available at: https://github.com/rohanmistry231/SQL-Interview-Preparation (Accessed: 29 May 2026).

MukundAabha (n.d.) *Data Science, ML, DL and NLP Interview Questions*. GitHub. Available at: https://github.com/MukundAabha/Data_Science_ML_DL_NLP_Interview_Qus/tree/main (Accessed: 29 May 2026).

Khosravi, V. (n.d.) *Interview Questions Answers: Non-technical Interview Questions*. GitHub. Available at: https://github.com/valakhosravi/Interview-Questions-Answers/tree/main/Non-technical (Accessed: 29 May 2026).

Information Commissioner's Office (ICO) (n.d.) *What is personal data?* Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/personal-information-what-is-it/what-is-personal-data/what-is-personal-data/ (Accessed: 29 May 2026).

Information Commissioner's Office (ICO) (n.d.) *A guide to the data protection principles*. Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/data-protection-principles/a-guide-to-the-data-protection-principles/ (Accessed: 29 May 2026).